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Editor-in-Chief
IEEE Transactions on Biometrics, Behavior, and Identity Science

Dear Editor-in-Chief,

I am submitting the manuscript "Manufactured Matches: Generative Face Restoration Inflates False Alarms in Open-Set Low-Resolution Recognition" for consideration as a regular paper in the IEEE Transactions on Biometrics, Behavior, and Identity Science.

Watchlist operators routinely receive low-resolution face probes and often enhance them with a super-resolution or face-restoration model before matching. The manuscript measures what this does to open-set identification with a frozen public encoder. It reports a security hazard that, to our knowledge, has not been quantified: restoration guided by a gallery candidate transfers that candidate's identity onto the probe, so a non-enrolled stranger is accepted as the candidate 6 to 14 times above the fixed 1% false-alarm baseline, with no gain in true detection. This failure mode cannot appear in verification or closed-set evaluation and is invisible to accuracy or rank-1, which is why the existing literature, that treats reference over-reliance as an image-fidelity artifact, has not surfaced it. The paper also establishes the resolution ceiling that explains the hazard and shows that only genuine multi-frame super-resolution crosses it, giving a concrete operating rule for deployment.

The results are obtained on public data (YouTube Faces, ChokePoint, TinyFace) with frozen public encoders (ArcFace and MagFace), and the hazard is confirmed across two encoders and across recording sessions. Every reported number is reproducible from the analysis scripts in the project repository.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All experiments use publicly available datasets under their respective terms. I am the sole author and the corresponding author. I have no conflicts of interest to declare.

Thank you for considering this submission.

Sincerely,

Alexander Apartsin